PG No. 37 $D_{3h}(1)$ $\bar{6}m2$ (-62m setting) [hexagonal]

Table 1: Representation matrices

Irrep.						
A_1'	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}1\end{bmatrix}$	$-6^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$-6^+_{001}:\begin{bmatrix}1\end{bmatrix}$
	$2_{110}:\begin{bmatrix}1\end{bmatrix}$	$2_{100}:\begin{bmatrix}1\end{bmatrix}$	$2_{010}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{120}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{210}:\begin{bmatrix}1\end{bmatrix}$
A_2'	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}1\end{bmatrix}$	$-6^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$-6^+_{001}:\begin{bmatrix}1\end{bmatrix}$
	$2_{110}:\begin{bmatrix}-1\end{bmatrix}$	$2_{100}:\begin{bmatrix}-1\end{bmatrix}$	$2_{010}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{120}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{210}:\begin{bmatrix}-1\end{bmatrix}$
E'	$1:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$-6^-_{001}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$-6^+_{001}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$
	$2_{110}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$2_{100}:\begin{bmatrix}1&0\\0&-1\end{bmatrix}$	$2_{010}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{120}:\begin{bmatrix}1&0\\0&-1\end{bmatrix}$	$\mathrm{m}_{210}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$
A_1''	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-6^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-6^+_{001}:\begin{bmatrix}-1\end{bmatrix}$
	$2_{110}:\begin{bmatrix}1\end{bmatrix}$	$2_{100}:\begin{bmatrix}1\end{bmatrix}$	$2_{010}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{120}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{210}:\begin{bmatrix}-1\end{bmatrix}$
A_2''	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-6^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-6^+_{001}:\begin{bmatrix}-1\end{bmatrix}$
	$2_{110}:\begin{bmatrix}-1\end{bmatrix}$	$2_{100}:\begin{bmatrix}-1\end{bmatrix}$	$2_{010}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{120}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{210}:\begin{bmatrix}1\end{bmatrix}$
E''	$1:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}-1&0\\0&-1\end{bmatrix}$	$-6^-_{001}:\begin{bmatrix}\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$-6^+_{001}:\begin{bmatrix}\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$
	$2_{110}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$2_{100}:\begin{bmatrix}1&0\\0&-1\end{bmatrix}$	$2_{010}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{120}:\begin{bmatrix}-1&0\\0&1\end{bmatrix}$	$\mathrm{m}_{210}:\begin{bmatrix}\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$